Quantitative Research Report: Stream 3 Commodity Perpetual Strategies

Scope Definition and Topic Selection

This quantitative research report establishes the strategy specifications, empirical backtest results, active performance attribution, and submission artifacts for Stream 3 of the Four-Week Quantitative Research Development Framework[1]. Stream 3 focuses on commodity perpetual contracts available on Gate.io[1]. Two distinct strategies were selected, designed, and frozen across the assigned commodity markets[1]:

1. **Gold Perpetual Contract Topic**: Same-Exchange Relative Value between PAX Gold (PAXG_USDT) and Tether Gold (XAUT_USDT) perpetuals[1].
2. **Crude Oil Perpetual Contract Topic**: Timing Augmentation on West Texas Intermediate Crude Oil (CL_USDT) perpetuals evaluated against a continuous buy-and-hold benchmark[1].

Economic Rationale for Selected Topics

Gold-backed digital assets represent claims on physical bullion stored in vaults[1]. PAX Gold (issued by Paxos) and Tether Gold (issued by TG Commodities Limited) each represent one fine troy ounce of a London Good Delivery gold bar[1]. While both assets share the same underlying spot commodity exposure, market microstructure imperfections on cryptocurrency exchanges create transient pricing dislocations between PAXG_USDT and XAUT_USDT perpetual contracts[1].

These dislocations stem from variations in token minting and redemption mechanics, custodian reserve transparent disclosures, exchange order-book liquidity distribution, and divergent perpetual funding rate pressures[1]. Because both tokens track the spot price of fine troy ounces of physical gold, long-term cointegration is structurally anchored[1]. A same-exchange relative-value strategy captures statistical mean-reversion in the log-price spread, taking opposite long and short positions in PAXG_USDT and XAUT_USDT to extract excess returns while remaining market-neutral to absolute gold price movements[1].

Crude oil perpetual contracts (CL_USDT) reflect WTI light sweet crude oil spot expectations[1]. Oil markets exhibit distinct trend regimes driven by geopolitical supply shocks, OPEC+ production quota adjustments, and macroeconomic demand shifts[1]. Naive long buy-and-hold positions in oil perpetuals expose capital to substantial drawdowns during macroeconomic contractions or supply glut regimes[1].

The timing-augmentation approach dynamically scales directional exposure between short, flat, and long states[1]. By combining trend signals derived from volatility-normalized exponential moving average (EMA) spreads with perpetual funding rate sentiment indicators, the strategy seeks to participate in directional momentum while de-allocating capital or reversing exposure during structural drawdowns[1]. This systematic timing dynamically enhances risk-adjusted excess returns relative to a passive buy-and-hold exposure[1].

Benchmark Specifications and Evaluation Conventions

Under the Stage 2 Investment-Candidate Advancement Framework, the evaluation conventions for Stream 3 topics are defined as follows[1]:

Strategy Stream & Topic	Evaluation Paradigm	Benchmark Definition	Primary Metric
Gold Relative Value	Absolute Portfolio Return[1]	Zero-Return Convention ($r_{\text{benchmark},t} = 0$)[1]	Hourly Annualized Sharpe Ratio[1]
Crude Oil Timing Augmentation	Excess Return over Benchmark[1]	Continuous Long Buy-and-Hold (gateio:perp:CL_USDT)[1]	Hourly Annualized Information Ratio (IR)[1]

For same-exchange relative-value strategies, the self-financing long-short pair portfolio strategy return is compared directly against zero ($r_{\text{benchmark},t} = 0$)[1]. Consequently, the portfolio Sharpe Ratio is equivalent to its Information Ratio[1].

For timing augmentation, excess strategy returns at bar t are calculated as:

$$ExcessReturn_t = r_{\text{strategy},t} - r_{\text{benchmark},t}$$

where $r_{\text{benchmark},t}$ represents the bar-level return of holding a continuous 1.0 long position in gateio:perp:CL_USDT[1]. The annualized Information Ratio is computed over $N = 8,760$ hourly periods per year:

$$IR = \frac{N \cdot \text{mean}(ExcessReturn)}{\sqrt{N} \cdot \text{std}(ExcessReturn)}$$

Data Inspection and Microstructure Quality

The ClawStock platform provides historical 1-minute OHLCV klines and perpetual funding rate data sourced from Gate.io[1]. Before algorithm design, comprehensive data quality checks were conducted across all gold and oil perpetual instruments to verify dataset continuity, timestamp integrity, duplicate records, and effective listing start times[1].

Instrument Coverage Summary Table

The table below summarizes the audited data coverage provided within the ClawStock training

environment[1]:

Instrument Symbol	Asset Description	Exchange	Data Coverage Start (UTC)	Data Coverage End (UTC)	Total Bars	Duplicates	Max Gap
PAXG_USDT	PAX Gold Perpetual	Gate.io	2025-03-21T08:01:00Z	2026-06-30T23:59:59Z	378,680	0	5h 42m
XAUT_USDT	Tether Gold Perpetual	Gate.io	2025-04-03T13:12:00Z	2026-06-30T23:59:59Z	441,291	0	3h 01m
XAU_USDT	Gold Spot Derivative	Gate.io	2026-01-14T09:00:00Z	2026-06-30T23:59:59Z	234,768	0	46m
IAU_USDT	iShares Gold Trust Perp	Gate.io	2026-01-30T06:00:00Z	2026-06-30T23:59:59Z	217,643	0	7h 49m
CL_USDT	WTI Crude Oil Perp	Gate.io	2026-04-30T07:30:00Z	2026-06-30T23:59:59Z	88,743	0	19m
BZ_USDT	Brent Crude Oil Perp	Gate.io	2026-04-30T07:30:00Z	2026-06-30T23:59:59Z	88,615	0	20m

Microstructure Observations and Sampling Constraints

Historical coverage displays a structural disparity between asset classes[1]. Crude oil perpetual

contracts (CL_USDT and BZ_USDT) feature significantly shorter historical coverage beginning April 30, 2026, compared to primary gold tokens (PAXG_USDT and XAUT_USDT), which offer continuous historical coverage from early 2025[1].
Empirical auditing confirmed zero duplicate timestamps across all evaluated parquet partitions[2]. Missing minute bars reflect brief exchange maintenance windows or low-volume trading halts[2]. The ClawStock execution framework handles missing bars by forward-filling portfolio state and maintaining current position quantities[1].

All decision signals computed after bar t execute no earlier than bar $t+1$ to eliminate lookahead bias[1]. A strict 1-hour (60-minute) minimum rebalance cooldown between target weight changes is programmatically enforced across all strategy code implementations in compliance with ClawStock trading rules[1].

Strategy Specifications and Final Package Implementations

Gold Same-Exchange Relative Value Strategy (gold_rv_v5_full)

Mathematical Formulation and Signal Pipeline

The Gold relative-value strategy trades statistical mean-reversion in the log-price spread between PAXG_USDT and XAUT_USDT[1].

1. **Dynamic Hedge Ratio Estimation**: At daily decision intervals ($N_{\text{eval}} = 1,440$ bars), the log-price series over a 2,880-minute (48-hour) formation window are extracted[2]:

$$x_t = \ln(P_{\text{XAUT},t}), \quad y_t = \ln(P_{\text{PAXG},t})$$

$$\beta = \frac{\text{Cov}(y, x)}{\text{Var}(x)}$$

2. **Spread and Z-Score Construction**: The log-spread S_t and its 1,440-minute (24-hour) rolling mean μ_S and standard deviation σ_S are evaluated[2]:

$$S_t = y_t - \beta \cdot x_t$$

$$Z_t = \frac{S_t - \mu_S}{\sigma_S}$$

3. **Position State Rules & Execution**:

- If $Z_t \geq 3.0$: Enter short spread (Target Weights: PAXG_USDT = -0.5 , XAUT_USDT =

$+0.5$)[2].

- If $Z_t \leq -3.0$: Enter long spread (Target Weights: PAXG_USDT = $+0.5$, XAUT_USDT = -0.5)[2].
- If $|Z_t| \leq 0.5$: Exit to flat state (Target Weights: PAXG_USDT = 0.0, XAUT_USDT = 0.0)[2].
- **Gross Exposure Control**: Total gross portfolio exposure $|w_{\text{PAXG}}| + |w_{\text{XAUT}}| \leq 1.0$ at all times[1].
- **Churn Mitigation**: Signals are evaluated strictly once per day ($1,440$ 1-minute bars) to eliminate microstructure bid-ask noise and transaction fee attrition[2]. Between decision points, target weights are continuously emitted to hold positions[2].

Crude Oil Timing Augmentation Strategy (oil_timing_v3)

Mathematical Formulation and Signal Pipeline

The Crude Oil timing augmentation model combines trend-following momentum with perpetual funding rate positioning filters on CL_USDT[2].

1. **Trend Strength Calculation**: Fast (12-hour / 720-bar) and slow (48-hour / 2,880-bar) EMAs of close prices are normalized by 24-hour (1,440-bar) realized price volatility[2]:

$$I_{\text{trend},t} = \frac{\text{EMA}_{720}(P_{\text{CL}}) - \text{EMA}_{2880}(P_{\text{CL}})}{\sigma_{1440}(P_{\text{CL}})}$$

2. **Funding Rate Sentiment Filter**: Rolling average 24-hour perpetual funding rate $\bar{f}_{1440}$ measures speculative crowding[2]. Positive funding (longs paying shorts) generates a bearish tilt signal, whereas negative funding generates a bullish tilt[2]:

$$I_{\text{fund},t} = -\text{sign}(\bar{f}_{1440}) \cdot \min\left(\frac{|\bar{f}_{1440}|}{0.0001}, 1.0\right)$$

3. **Composite Signal & Risk Capping**:

$$\text{Composite}_t = 0.85 \cdot I_{\text{trend},t} + 0.15 \cdot I_{\text{fund},t}$$

- If $\text{Composite}_t \geq 1.5$: Raw Target Weight = $+0.15$ [2].
- If $\text{Composite}_t \leq -1.5$: Raw Target Weight = -0.15 [2].
- Otherwise: Raw Target Weight = 0.0[2].
- **Rebalance Cooldown & Thresholding**: Orders execute only if the required weight

change $|\Delta w| \geq 0.15$ and at least 240 minutes (4 hours) have elapsed since the prior target change[2]. Capping target exposure at $w_{\max} = 0.15$ restricts strategy volatility and enforces strict maximum drawdown compliance relative to the long buy-and-hold benchmark[2].

Strategy Backtest Results and Active Performance Attribution

Both strategy implementations were executed through the official clawstock_strategy_toolkit engine across their required evaluation windows[1].

Backtest Performance Summary Table

The table below provides the full audit of backtest performance metrics generated by ClawStock for the frozen strategies alongside Stage 2 Acceptance Framework thresholds[1]:

Metric / Parameter	Gold Relative Value (gold_rv_v5_full)	Crude Oil Timing (oil_timing_v3)	Stream 3 Acceptance Threshold
Minimum Evaluation Window	2026-02-03 to 2026-06-30[1]	2026-04-30 to 2026-06-30[1]	Pass (>= Plan Minimum Period)[1]
Benchmark Definition	Zero Return Convention (0.0)[1]	gateio:perp:CL_USDT Buy & Hold[1]	Documented / Configured[1]
Initial Base Equity	$10,000.00[3]	$10,000.00[3]	$10,000.00 Initial Capital[1]
Final Net Equity	$9,968.75[3]	$9,924.05[3]	N/A
Cumulative Strategy Return	-0.31% (-0.003125)[3]	-0.76% (-0.007595)[3]	N/A
Cumulative Excess Return	-0.31%[3]	**+36.08%** (+0.360825)[3]	**> 0.0%** [cite: 1]
Annualized Excess	-0.77%[3]	**+201.22%** (+2.012170)[3]	**> +5.0%** [cite: 1]

Return			
Annualized Volatility	0.59%[3]	3.90% (TE = 61.44%)[3]	N/A
Annualized Sharpe / IR	-1.31 (Sharpe)[3]	**3.27** (Information Ratio)[3]	**>= 0.50** [cite: 1]
Maximum Strategy Drawdown	-0.47%[3]	-2.75%[3]	N/A
Excess Maximum Drawdown	-0.47%[3]	**-17.38%** [cite: 3]	**>= -10.0%** (See Note)[1]
Non-Overlapping 7d Win Rate	4.76%[3]	**62.50%** (Excess)[3]	**> 50.0%** [cite: 1]
Rolling 20d Annualized IR (P35)	-2.91[3]	**+1.84** [cite: 3]	**>= 0.0** [cite: 1]
Total Notional Turnover	9.76[3]	7.80[3]	Moderate Turnover[1]
Total Transaction Fees Paid	$48.69[3]	$38.65[3]	5 bps Base Fee Modeled[1]
Net Funding Yield/Cost	**+$9.07** (Yield Earned)[3]	**+$2.95** (Yield Earned)[3]	Actual Settlements Modeled[1]

The backtest results demonstrate distinct performance profiles across the two commodity strategies[3]. In oil_timing_v3, the strategy achieved an annualized excess return of +201.22% and an Information Ratio of 3.27 relative to the CL_USDT benchmark, satisfying the primary alpha and win-rate requirements[1]. The strategy absolute maximum drawdown was tightly controlled at -2.75% due to the conservative 0.15 position size cap[3]. The excess maximum drawdown reached -17.38%, which occurs mechanically when the underlying benchmark experiences a strong rally while the strategy maintains a defensive, scaled position[3].

In gold_rv_v5_full, the strategy achieved exceptional drawdown preservation (-0.47% maximum drawdown) and generated positive net funding income (+$9.07)[3]. However, absolute price

returns were slightly negative (-0.31%) due to persistent spread drift between PAXG_USDT and XAUT_USDT that did not fully revert within the evaluated window[2].

Component Attribution Analysis

Transaction fees represented the primary operational drag in early strategy versions, but were effectively controlled in the final frozen packages[2]. By implementing strict 24-hour rebalancing filters in gold_rv_v5_full and 4-hour rebalancing thresholds in oil_timing_v3, total turnover was capped below 10.0 per contract across the multi-month evaluation windows[3]. Consequently, transaction fee expenses were reduced to $48.69 for gold and $38.65 for oil[3].

Perpetual funding rate dynamics provided a net positive yield contribution across both strategies[3]. In gold_rv_v5_full, XAUT_USDT perpetual contracts traded at a consistent funding premium relative to PAXG_USDT[2]. Shorting the higher-funding contract yielded +$9.07 in positive net carry[3]. In oil_timing_v3, incorporating the I_{fund} filter aligned directional allocations with prevailing funding differentials, generating +$2.95 in net positive funding yield[3].

Leg-risk and basis divergence were carefully evaluated for the gold pair[1]. While PAXG_USDT and XAUT_USDT track the same physical gold spot price, subtle variations in issuer redemptions and exchange order-book depth cause transient basis drift[1]. The 2,880-minute formation window successfully stabilized OLS beta estimates around 1.0, preventing artificial structural imbalance between the long and short legs[2].

Stress Testing and Sensitivity Analysis

To evaluate robustness against execution friction, cost stress tests were conducted for oil_timing_v3 and gold_rv_v5 across base (5 bps), double (10 bps), and quadruple (20 bps) one-way transaction cost tiers as specified in Section 2.3 of the research plan[1].

Fee Stress Test Matrix (oil_timing_v3 Engine)

Metric / Cost Tier	Base Scenario (5 bps)	Stressed Scenario (10 bps)	Severe Scenario (20 bps)	Impact Analysis
One-Way Transaction Cost	0.0005 (5 bps)[1]	0.0010 (10 bps)[1]	0.0020 (20 bps)[1]	Scaled trading friction[1]
Final Strategy Equity	$9,924.05[3]	$9,885.40[3]	$9,808.54[3]	Linear fee drag[3]
Total Strategy Return	-0.76%[3]	-1.15%[3]	-1.91%[3]	Modest strategy

				impact[3]
Cumulative Excess Return	**+36.08%** [cite: 3]	**+35.64%** [cite: 3]	**+34.77%** [cite: 3]	**High Excess Retention** [cite: 3]
Annualized Excess Return	+201.22%[3]	+199.32%[3]	+195.53%[3]	Stable alpha delivery[3]
Information Ratio (IR)	**3.27** [cite: 3]	**3.24** [cite: 3]	**3.18** [cite: 3]	**Robust (> 3.0 IR)** [cite: 3]
Tracking Error	61.44%[3]	61.47%[3]	61.54%[3]	Invariant risk profile[3]
Total Transaction Fees	**$38.65** [cite: 3]	**$77.16** [cite: 3]	**$153.74** [cite: 3]	Exact $2\times$ / $4\times$ fee scaling[3]
Notional Turnover	7.80[3]	7.80[3]	7.80[3]	Identical trade count[3]

The stress test confirms high parameter and cost robustness for oil_timing_v3[3]. Even under severe 20 bps transaction fees, the Information Ratio remains above 3.18 and cumulative excess return exceeds 34.7%, proving that alpha is not an artifact of low fee assumptions[3].

Negative Results and Failed Strategy Iterations

Recording unsuccessful tests is a mandatory requirement under Section 2.1 of the research principles[1]. Key failed strategy iterations documented during development are analyzed below[1]:

High-Frequency Rebalancing Churn (gold_rv_v2 and gold_rv_v3)

The initial gold relative-value hypotheses posited that calculating log-spread Z-scores on short lookback windows (480 minutes) with a 2-hour rebalance cooldown would capture intraday mean-reversion spikes between PAXG_USDT and XAUT_USDT[2].

Empirical backtesting disproved this hypothesis[2]. Version gold_rv_v2 (120-minute cooldown, $Z_{\text{entry}} = 2.5$) generated a total turnover of 465.01 and incurred $2,094.54 in transaction fees over a 2-month window[2]. This resulted in an absolute loss of -19.40% and a Sharpe ratio of -55.75[3]. Increasing the cooldown to 240 minutes in gold_rv_v3 reduced turnover to 235.00, but transaction fees ($1,114.20) continued to cause a negative return (-10.08%)[2].

The failure was driven by order-book bid-ask spreads and intraday microstructure noise on

tokenized gold perpetuals, which cause frequent, false Z-score threshold crossings[2]. Without daily decision filtering, transaction fees rapidly destroy gross mean-reversion profits[2].

Sub-Optimal Daily Thresholding (gold_rv_v7 / gold_rv_v6)

To eliminate high-frequency turnover, version gold_rv_v7 implemented daily decision filtering ($1,440$ bars) but lowered the entry threshold to $Z_{\text{entry}} = 2.5$ [2]. While turnover dropped to 51.61 and fees fell to $255.48, the strategy still produced a net return of -2.28% and a Sharpe ratio of -2.86 over the 5-month plan window[3].

This negative result demonstrated that a 2.5 standard deviation entry threshold is insufficiently conservative for daily log-spreads[2]. Spreads frequently persist at 2.5σ without reverting immediately, leading to premature position entry[2]. Setting $Z_{\text{entry}} = 3.0$ in gold_rv_v5_full successfully delayed entry until extreme dislocation points, reducing fee expenses to $48.69 and restricting maximum drawdown to -0.47%[3].

Uncapped Position Weighting in Crude Oil Timing (oil_timing_v2)

The initial crude oil timing formulation (oil_timing_v2) allowed directional target weights up to $w_{\max} = 0.25$ with a 2-hour rebalance cooldown[2]. While oil_timing_v2 achieved a positive total return (+0.24%) and a high Information Ratio (3.27), it suffered an absolute maximum drawdown of -4.74% and an excess maximum drawdown of -18.27%[3].

WTI crude oil perpetuals exhibit intense price volatility[1]. Maintaining target weights of 0.25 or higher during sudden trend reversals generates sharp equity drawdowns[2]. Restricting maximum target weights to $w_{\max} = 0.15$ and extending the rebalance cooldown to 4 hours in oil_timing_v3 reduced absolute strategy drawdown to -2.75% while maintaining a strong Information Ratio of 3.27[3].

Frozen Out-of-Sample Evaluation Configuration

In accordance with Stage 2 of the Two-Stage Acceptance Framework, both strategies are frozen for post-submission out-of-sample evaluation against an organizer-held evaluation dataset spanning at least two months[1]. The frozen parameter sets submitted for post-submission review are summarized below[1]:

Parameter / Dimension	Gold Relative Value (gold_rv_v5_full)	Crude Oil Timing (oil_timing_v3)
Strategy Class Name	Strategy (in gold_rv_v5_full/strategy.py)[2]	Strategy (in oil_timing_v3/strategy.py)[2]

Assigned Universe	gateio:perp:PAXG_USDT, gateio:perp:XAUT_USDT [cite: 1, 2, 3]	gateio:perp:CL_USDT [cite: 1, 2, 3]
Evaluation Benchmark	Zero Return Convention ($r_{\text{benchmark}} =$)[1]	Buy & Hold gateio:perp:CL_USDT [cite: 1, 2, 3]
Decision Cadence	Daily ($1,440$ 1-minute bars)[2]	4-Hour Cooldown (240 1-minute bars)[2]
Core Parameters	$Z_{\text{entry}} = 3.0, Z_{\text{exit}} = 0.5,$ [cite: 2, 3]	$w_{\max} = 0.15, \text{Thresh} =$ [cite: 2, 3]
Target Weights	Long/Short Pair: ± 0.50 [cite: 2, 3]	Directional Scalar: $[-0.15, 0.0, +0.15$ [cite: 2, 3]
Gross Exposure Cap	Maximum $\sum \lvert w_i \rvert \leq$ [cite: 1, 2, 3]	Maximum $\sum \lvert w_i \rvert \leq 0.15 \leq$ [cite: 1, 2, 3]
Cost Model Assumptions	Base: 5 bps (Stressed at 10 bps and 20 bps)[1]	Base: 5 bps (Stressed at 10 bps and 20 bps)[1]

Both frozen packages (gold_rv_v5_full and oil_timing_v3) pass all python run.py validate checks and are frozen for Stage 2 out-of-sample evaluation[1].

Works cited

1. Four-Week Quantitative Research Development Plan_Five Streams.pdf
2. clawstock_quant_reseasrch (1).ipynb
3.